

- 3 Prove the identity $\frac{\cos x}{\operatorname{cosec} x - 1} + \frac{\cos x}{\operatorname{cosec} x + 1} = 2 \tan x$. [5]

$$\text{LHS} = \frac{\cos x}{\operatorname{cosec} x - 1} + \frac{\cos x}{\operatorname{cosec} x + 1}$$

$$= \frac{\cos x (\operatorname{cosec} x + 1) + \cos x (\operatorname{cosec} x - 1)}{(\operatorname{cosec} x)^2 - (1)^2} \quad (M1)$$

$$= \frac{\cos x \left[\frac{1}{\sin x} + 1 \right] + \cos x \left[\frac{1}{\sin x} - 1 \right]}{\operatorname{cosec}^2 x - 1} \quad (M1)$$

$$| 1 + \cot^2 x = \operatorname{cosec}^2 x |$$

$$= \frac{\frac{\cos x}{\sin x} + \cancel{\cos x} + \frac{\cos x}{\sin x} - \cancel{\cos x}}{\cot^2 x}$$

$$\frac{\cos x}{\sin x} = \cot x$$

$$= \frac{2 \cot x}{\cot^2 x} \quad (M1)$$

$$= \frac{2}{\cot x} \quad (M1)$$

$$= 2 \tan x$$

$$= \text{RHS (shown)} \quad (A1)$$